

Does Optimization Parallel
Inference Change LLM
Inference Outputs?

Motivation

- **Modern LLM inference systems rely heavily on Optimization Parallel Inference.**
- **These techniques significantly improve throughput and latency, but they rely on an implicit assumption**
 - **Given the same model and input, different system execution paths should produce the almost same output.**

Problem Statement

For the same model and input, do different system-level inference configurations produce similar outputs?

- Does KV cache location (local vs remote) matter?
- Does asynchronous communication change results?
- Does decoupled prefill/decode execution alter outputs?

Approach

We use a controlled experimental setup:

- Fixed: model weights, inputs, decoding strategy
- Varied system settings:
 - Local vs remote KV cache
 - Synchronous vs asynchronous communication
 - Monolithic vs prefill/decode-decoupled execution

Expected Outcome

Possible outcomes:

- (1) Confirmation that outputs are invariant under common system optimizations
- (2) Identification of system conditions that introduce structured output differences